Clustering Premier League Teams by Playing Style

ECS784P/U Data Analytics — Coursework 1

Module: ECS784P/U Data Analytics
Institution: Queen Mary University of London
Dataset: English Premier League 2024–2025 Season (FBref)
Methods: K-Means Clustering & Gaussian Mixture Models (GMM)

All statistics are per-90-minute squad averages. Analysis covers all 20 EPL teams across 15 playing-style features.

Overview

Objective

Our objective is to cluster English Premier League Teams on their playing style, explore similarities using two **Unsupervised Learning techniques: GMM and K-Means**.

1. Imports & Data Loading

In [6]:
```
# Data manipulation
import pandas as pd

# Visualisation
import matplotlib.pyplot as plt
import seaborn as sns
```

In [7]:
```
# Load the pre-merged master dataset (produced by Data_Analytics_Da
# 20 EPL squads × 47 raw columns; all statistics are per-90-minute
data = pd.read_csv('epl_2024_2025_master_stats.csv')
```

2. Exploratory Data Analysis (EDA)

We begin by inspecting the dataset's shape, column types, and missing values, then visualise feature correlations and use PCA to understand the intrinsic dimensionality before clustering.

Shape, Columns

In [8]:
```
print("Shape:", data.shape)
print("\nColumns:", data.columns.tolist())
data.head()
```

```
Shape: (20, 47)

Columns: ['Squad', '# Pl', 'Age', 'Poss', 'MP', 'Starts', 'Min', '90
s', 'Gls', 'Ast', 'G+A', 'G-PK', 'PK', 'PKatt', 'CrdY', 'CrdR', 'G+A
-PK', 'Sh', 'SoT', 'SoT%', 'Sh/90', 'SoT/90', 'G/Sh', 'G/SoT', '2Crd
Y', 'Fls', 'Fld', 'Off', 'Crs', 'Int', 'TklW', 'PKwon', 'PKcon', 'O
G', 'GA', 'GA90', 'SoTA', 'Saves', 'Save%', 'W', 'D', 'L', 'CS', 'C
S%', 'PKA', 'PKsv', 'PKm']
```

Out[8]:

	Squad	# Pl	Age	Poss	MP	Starts	Min	90s	Gls	Ast	...	Sa
0	Arsenal	25	25.8	56.9	38	418	3,420	38.0	1.76	1.45	...	2
1	Aston Villa	28	27.0	50.5	38	418	3,420	38.0	1.47	1.18	...	2
2	Bournemouth	29	25.1	48.5	38	418	3,420	38.0	1.50	1.08	...	3
3	Brentford	28	25.8	47.9	38	418	3,420	38.0	1.71	1.16	...	4
4	Brighton	32	24.8	52.3	38	418	3,420	38.0	1.68	1.08	...	2

5 rows × 47 columns

Check for missing values and data types

In [9]:
```
print(data.dtypes)
print("\nMissing values:")
print(data.isnull().sum())
```

```
Squad      object
# Pl        int64
Age       float64
Poss      float64
MP          int64
Starts      int64
Min        object
90s       float64
Gls       float64
Ast       float64
G+A       float64
G-PK      float64
PK        float64
PKatt     float64
CrdY      float64
CrdR      float64
G+A-PK    float64
Sh        float64
SoT       float64
SoT%      float64
Sh/90     float64
```

```
SoT/90    float64
G/Sh      float64
G/SoT     float64
2CrdY     float64
Fls       float64
Fld       float64
Off       float64
Crs       float64
Int       float64
TklW      float64
PKwon     float64
PKcon     float64
OG        float64
GA        float64
GA90      float64
SoTA      float64
Saves     float64
Save%     float64
W         float64
D         float64
L         float64
CS        float64
CS%       float64
PKA       float64
PKsv      float64
PKm       float64
dtype: object

Missing values:
Squad     0
# Pl      0
Age       0
Poss      0
MP        0
Starts    0
Min       0
90s       0
Gls       0
Ast       0
G+A       0
G-PK      0
PK        0
PKatt     0
CrdY      0
CrdR      0
G+A-PK    0
Sh        0
SoT       0
SoT%      0
Sh/90     0
SoT/90    0
G/Sh      0
G/SoT     0
2CrdY     0
Fls       0
Fld       0
```

```
Off       0
Crs       0
Int       0
TklW      0
PKwon     0
PKcon     0
OG        0
GA        0
GA90      0
SoTA      0
Saves     0
Save%     0
W         0
D         0
L         0
CS        0
CS%       0
PKA       0
PKsv      0
PKm       0
dtype: int64
```

3. Feature Selection

We filter the 47 raw columns to retain only features that describe *how* a team plays, not *how well* they performed.

Observations

The dataset is clean — no missing values and all columns are numeric (except `Squad` and `Min`).

We now identify and drop features that do not describe *how* a team plays:

- **Metadata** (`# Pl`, `Age`, `MP`, `Starts`, `Min`, `90s`): squad composition or scheduling constants, identical across teams
- **Outcome metrics** (`Gls`, `Ast`, `W`, `D`, `L`, `CS`, `G/Sh`, `G/SoT`, `GA`, `GA90`, etc.): measure results, not the manner of play
- **Rare events** (`PKwon`, `PKcon`, `2CrdY`, `OG`): too infrequent to reliably characterise a playing style

In [10]:
```
# Metadata columns (not style-related)
# Outcome columns (measure results, not how the team plays)

cols_to_drop = [
    '# Pl',        # squad size — not style
    'Age',         # average age — not style
    'MP',          # matches played — same for all
    'Starts',      # same for all
    'Min',         # same for all
    '90s',         # same for all
```

```
    'Gls',          # goals scored – outcome
    'Ast',          # assists – outcome
    'G+A',          # goals + assists – outcome
    'G-PK',         # non-penalty goals – outcome
    'PK',           # penalties scored – outcome
    'PKatt',        # penalties attempted – outcome
    'G+A-PK',       # outcome
    'G/Sh',         # goals per shot – outcome (finishing quality, no
    'G/SoT',        # goals per shot on target – outcome
    'GA',           # goals against – outcome
    'GA90',         # goals against per 90 – outcome
    'W',            # wins – outcome
    'D',            # draws – outcome
    'L',            # losses – outcome
    'CS',           # clean sheets – outcome
    'CS%',          # clean sheet % – outcome
    'PKwon',        # penalty events – too rare to define style
    'PKcon',        # penalty events
    'PKA',          # penalty events
    'PKsv',         # penalty events
    'PKm',          # penalty events
    'OG',           # own goals – random events
    '2CrdY',        # second yellows – too rare
]

data_style = data.drop(columns=cols_to_drop)

# Set Squad as index
data_style = data_style.set_index('Squad')

print("Shape:", data_style.shape)
data_style.columns.tolist()
```

```
Shape: (20, 17)
```

Out[10]:

```
['Poss',
 'CrdY',
 'CrdR',
 'Sh',
 'SoT',
 'SoT%',
 'Sh/90',
 'SoT/90',
 'Fls',
 'Fld',
 'Off',
 'Crs',
 'Int',
 'TklW',
 'SoTA',
 'Saves',
 'Save%']
```

3. Data Preprocessing

3.1 Feature Scaling (StandardScaler)

K-Means and GMM both rely on distance or covariance metrics. Without scaling, features with large numeric ranges (e.g. `Crs` ≈ 14–22 per 90) would dominate over features with small ranges (e.g. `CrdR` ≈ 0.03–0.13 per 90), regardless of statistical importance.

`StandardScaler` transforms each feature to **zero mean and unit standard deviation**, ensuring all 15 features contribute equally to the distance metric.

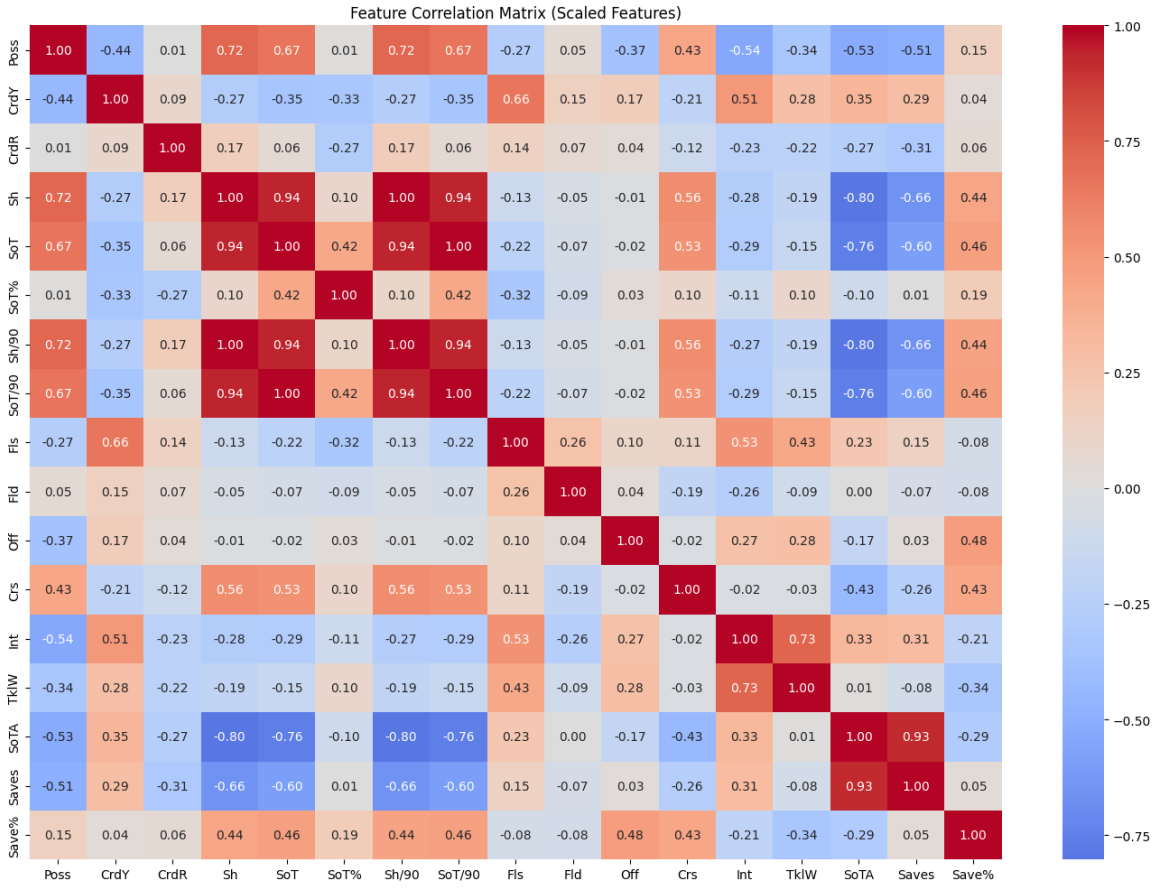
In [11]:
```
from sklearn.preprocessing import StandardScaler

# StandardScaler: transforms each feature to mean=0, std=1
# Essential for K-Means (Euclidean distance) and GMM (covariance-ba
# feature dominates purely due to its scale rather than its discrim
scaler = StandardScaler()
data_scaled = pd.DataFrame(
    scaler.fit_transform(data_style),
    columns=data_style.columns,
    index=data_style.index
)

data_scaled.head()
```

Out[11]:

Squad	Poss	CrdY	CrdR	Sh	SoT	Sc
Arsenal	1.194296	-0.900957	2.450526	0.669885	0.453202	-0.401
Aston Villa	0.087345	-0.351802	1.104083	-0.114146	-0.179762	-0.133
Bournemouth	-0.258577	1.535918	0.296217	1.038841	0.921596	-0.089
Brentford	-0.362353	-1.621723	-1.050225	-0.667579	0.288632	2.812
Brighton	0.398675	-0.180191	0.296217	0.439288	0.630432	0.535

Check correlations before PCA

In [12]:
```
import os
# Ensure output directory exists for saving report figures
os.makedirs('dist/figures', exist_ok=True)

plt.figure(figsize=(14, 10))
sns.heatmap(data_scaled.corr(), annot=True, fmt='.2f', cmap='coolwa
plt.title('Feature Correlation Matrix (Scaled Features)')
plt.tight_layout()
plt.savefig('dist/figures/fig_corr_heatmap.png', dpi=150, bbox_inch
plt.show()
```



In [13]:
```
data_scaled.corr()
```

Out[13]:

	Poss	CrdY	CrdR	Sh	SoT	SoT%
Poss	1.000000	-0.441869	0.006032	0.723381	0.668081	0.013244
CrdY	-0.441869	1.000000	0.087573	-0.274635	-0.349094	-0.328718
CrdR	0.006032	0.087573	1.000000	0.168439	0.059896	-0.274742
Sh	0.723381	-0.274635	0.168439	1.000000	0.941943	0.097968
SoT	0.668081	-0.349094	0.059896	0.941943	1.000000	0.424238
SoT%	0.013244	-0.328718	-0.274742	0.097968	0.424238	1.000000
Sh/90	0.722639	-0.271807	0.169735	0.999928	0.942357	0.099137
SoT/90	0.668081	-0.349094	0.059896	0.941943	1.000000	0.424238
Fls	-0.269038	0.660632	0.140953	-0.129990	-0.219801	-0.323493
Fld	0.051699	0.150697	0.069617	-0.051549	-0.071670	-0.086803
Off	-0.371987	0.173693	0.044979	-0.014599	-0.015929	0.029265
Crs	0.432357	-0.207232	-0.124374	0.559026	0.532407	0.096353
Int	-0.540547	0.505718	-0.227360	-0.275708	-0.287705	-0.112153
TklW	-0.340730	0.275844	-0.223029	-0.194072	-0.151847	0.096045
SoTA	-0.526170	0.347098	-0.266966	-0.802579	-0.757254	-0.102066
Saves	-0.506299	0.291656	-0.311108	-0.660098	-0.596121	0.005848
Save%	0.153499	0.038664	0.062051	0.441242	0.461658	0.193509

Key observation from the correlation matrix:

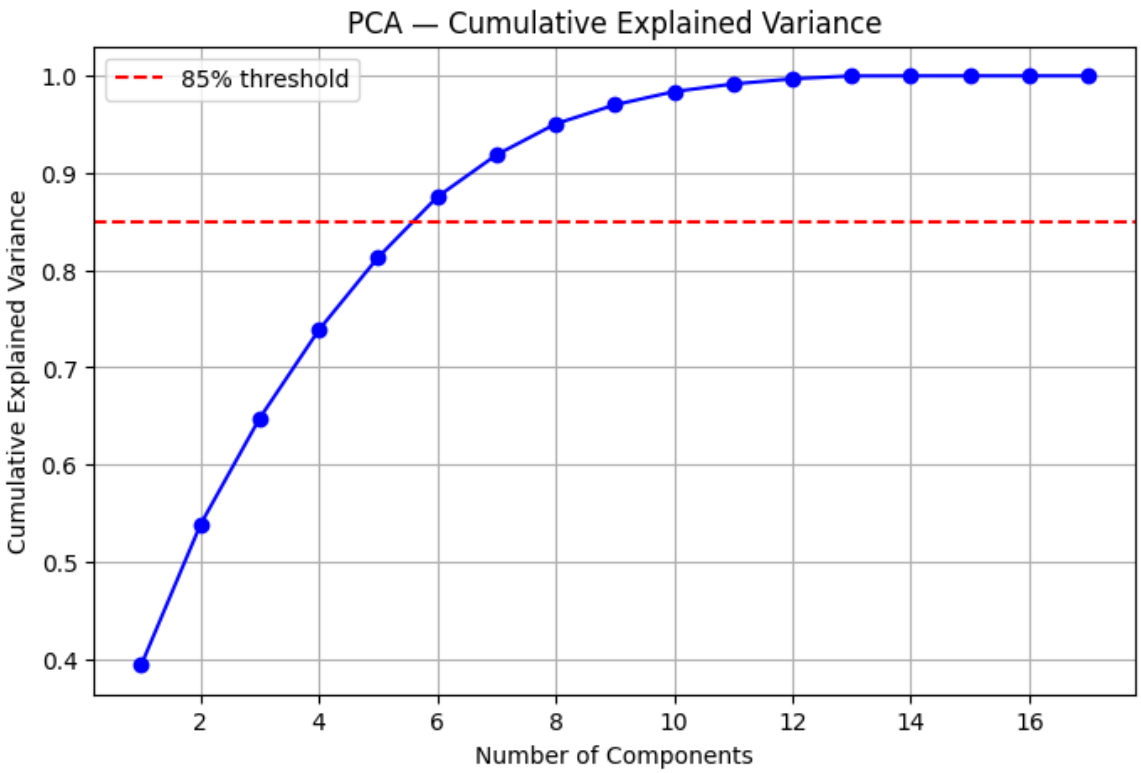
- `Sh` and `Sh/90` have a near-perfect correlation of $r = 0.9999$
- `SoT` and `SoT/90` are identical ($r = 1.0$)

This is expected: since all 20 EPL teams played exactly 38 matches, per-90 and total values are directly proportional. Retaining both would **double-weight** those features in the distance metric. We drop `Sh/90` and `SoT/90` before re-scaling.

PCA to understand dimensionality

In [14]:
```
from sklearn.decomposition import PCA

# Fit PCA with all 15 components to assess intrinsic dimensionality
# PCA is used for EXPLORATION ONLY — clustering is performed on all
pca_full = PCA()
pca_full.fit(data_scaled)

plt.figure(figsize=(8, 5))
```

```
plt.plot(range(1, len(pca_full.explained_variance_ratio_) + 1),
         pca_full.explained_variance_ratio_.cumsum(), 'bo-')
plt.xlabel('Number of Components')
plt.ylabel('Cumulative Explained Variance')
plt.title('PCA — Cumulative Explained Variance')
plt.axhline(y=0.85, color='r', linestyle='--', label='85% threshold
plt.legend()
plt.grid(True)
plt.savefig('dist/figures/fig_pca_variance.png', dpi=150, bbox_inch
plt.show()

print("Variance explained per component:")
for i, var in enumerate(pca_full.explained_variance_ratio_):
    print(f"  PC{i+1}: {var:.3f} (cumulative: {sum(pca_full.explain
```




```
Variance explained per component:
  PC1: 0.394 (cumulative: 0.394)
  PC2: 0.144 (cumulative: 0.538)
  PC3: 0.109 (cumulative: 0.648)
  PC4: 0.091 (cumulative: 0.739)
  PC5: 0.075 (cumulative: 0.813)
  PC6: 0.062 (cumulative: 0.875)
  PC7: 0.043 (cumulative: 0.919)
  PC8: 0.032 (cumulative: 0.951)
  PC9: 0.020 (cumulative: 0.970)
  PC10: 0.014 (cumulative: 0.984)
  PC11: 0.008 (cumulative: 0.992)
  PC12: 0.005 (cumulative: 0.997)
  PC13: 0.003 (cumulative: 1.000)
  PC14: 0.000 (cumulative: 1.000)
  PC15: 0.000 (cumulative: 1.000)
  PC16: 0.000 (cumulative: 1.000)
  PC17: 0.000 (cumulative: 1.000)
```

Correlation matrix: Sh and Sh/90 are almost perfectly correlated (0.999),

and SoT and SoT/90 are identical (1.0). This makes sense — with all teams playing 38 matches, per-match and per-90 are the same thing. Similarly, SoTA and Saves are very highly correlated (0.93). We should drop the redundant columns before clustering to avoid giving those features double weight.

In [15]:
```
# Sh/90 is identical to Sh, SoT/90 is identical to SoT (same matche
data_style = data_style.drop(columns=['Sh/90', 'SoT/90'])

print("Shape after removing redundant features:", data_style.shape)
print("Remaining columns:", data_style.columns.tolist())
```

```
Shape after removing redundant features: (20, 15)
Remaining columns: ['Poss', 'CrdY', 'CrdR', 'Sh', 'SoT', 'SoT%', 'Fl
s', 'Fld', 'Off', 'Crs', 'Int', 'TklW', 'SoTA', 'Saves', 'Save%']
```

Re-scale and re-run PCA

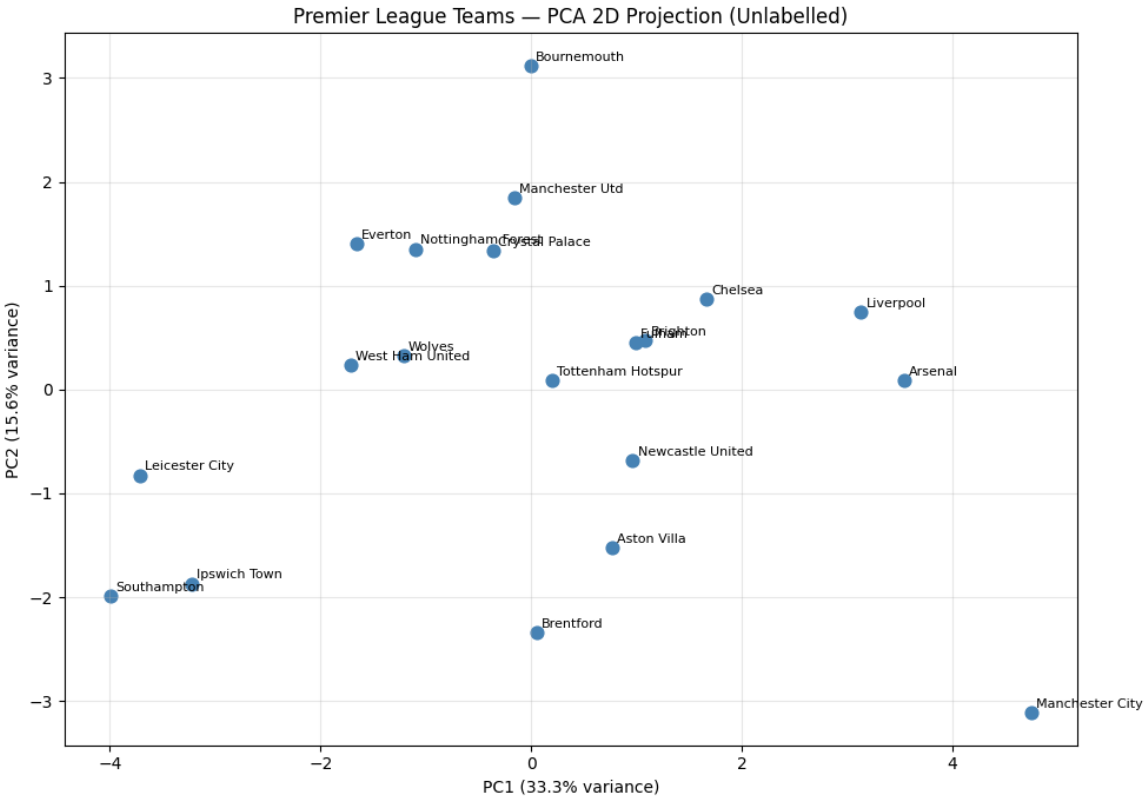
In [16]:
```
data_scaled = pd.DataFrame(
    scaler.fit_transform(data_style),
    columns=data_style.columns,
    index=data_style.index
)

pca_full = PCA()
pca_full.fit(data_scaled)

print("Variance explained per component:")
for i, var in enumerate(pca_full.explained_variance_ratio_):
    print(f"  PC{i+1}: {var:.3f} (cumulative: {sum(pca_full.explain
```

```
Variance explained per component:
  PC1: 0.333 (cumulative: 0.333)
  PC2: 0.156 (cumulative: 0.489)
  PC3: 0.122 (cumulative: 0.611)
  PC4: 0.103 (cumulative: 0.714)
  PC5: 0.083 (cumulative: 0.798)
  PC6: 0.070 (cumulative: 0.867)
  PC7: 0.046 (cumulative: 0.913)
  PC8: 0.033 (cumulative: 0.947)
  PC9: 0.021 (cumulative: 0.967)
  PC10: 0.015 (cumulative: 0.982)
  PC11: 0.009 (cumulative: 0.991)
  PC12: 0.006 (cumulative: 0.997)
  PC13: 0.003 (cumulative: 1.000)
  PC14: 0.000 (cumulative: 1.000)
  PC15: 0.000 (cumulative: 1.000)
```

Apply PCA with chosen number of components and visualize in 2D

In [17]:
```
# Project onto first 2 PCs for 2D visualisation (exploration only —
pca_2d = PCA(n_components=2)
data_2d = pca_2d.fit_transform(data_scaled)
```

```
plt.figure(figsize=(10, 7))
plt.scatter(data_2d[:, 0], data_2d[:, 1], color='steelblue', s=60)

for i, team in enumerate(data_style.index):
    plt.annotate(team, (data_2d[i, 0], data_2d[i, 1]), fontsize=8,
                 xytext=(3, 3), textcoords='offset points')

plt.xlabel(f'PC1 ({pca_2d.explained_variance_ratio_[0]:.1%} varianc
plt.ylabel(f'PC2 ({pca_2d.explained_variance_ratio_[1]:.1%} varianc
plt.title('Premier League Teams — PCA 2D Projection (Unlabelled)')
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('dist/figures/fig_pca_2d_unlabelled.png', dpi=150, bbox_
plt.show()
```



4. Clustering with K-Means and GMM

We apply two unsupervised learning algorithms to the **20 × 15 scaled feature matrix**:

- **K-Means** — hard cluster assignments; minimises within-cluster sum of squares (inertia)
- **GMM** — probabilistic soft assignments; models each cluster as a multivariate Gaussian fitted via the EM algorithm

Both methods require the number of clusters k to be specified in advance. We select k using:

- **Elbow method** (K-Means) — identify where inertia reduction flattens

- **Bayesian Information Criterion / BIC** (GMM) — lower BIC indicates better fit with fewer parameters

In [18]:
```
from sklearn.cluster import KMeans
from sklearn.mixture import GaussianMixture

# Evaluate k = 1 to 10 to determine the optimal number of clusters
kmeans_scores = []  # inertia (within-cluster sum of squares) — low
gmm_scores = []     # BIC score — lower is better, penalises model

for k in range(1, 11):
    # K-Means: k-means++ initialisation with 10 restarts avoids poo
    km = KMeans(n_clusters=k, init='k-means++', max_iter=200, n_ini
    km.fit(data_scaled)
    kmeans_scores.append(km.inertia_)

    # GMM: 10 initialisations for numerical stability
    gm = GaussianMixture(n_components=k, random_state=123, n_init=1
    gm.fit(data_scaled)
    gmm_scores.append(gm.bic(data_scaled))

    print(f"k={k}: KMeans inertia={km.inertia_:.2f}, GMM BIC={gm.bi
```

```
k=1: KMeans inertia=300.00, GMM BIC=809.27
k=2: KMeans inertia=233.14, GMM BIC=-178.86
k=3: KMeans inertia=192.95, GMM BIC=-596.04
k=4: KMeans inertia=163.64, GMM BIC=-657.32
k=5: KMeans inertia=140.77, GMM BIC=-367.54
k=6: KMeans inertia=128.54, GMM BIC=-217.81
k=7: KMeans inertia=104.50, GMM BIC=9.87
k=8: KMeans inertia=88.91, GMM BIC=363.32
k=9: KMeans inertia=73.93, GMM BIC=651.05
k=10: KMeans inertia=69.81, GMM BIC=972.49
```

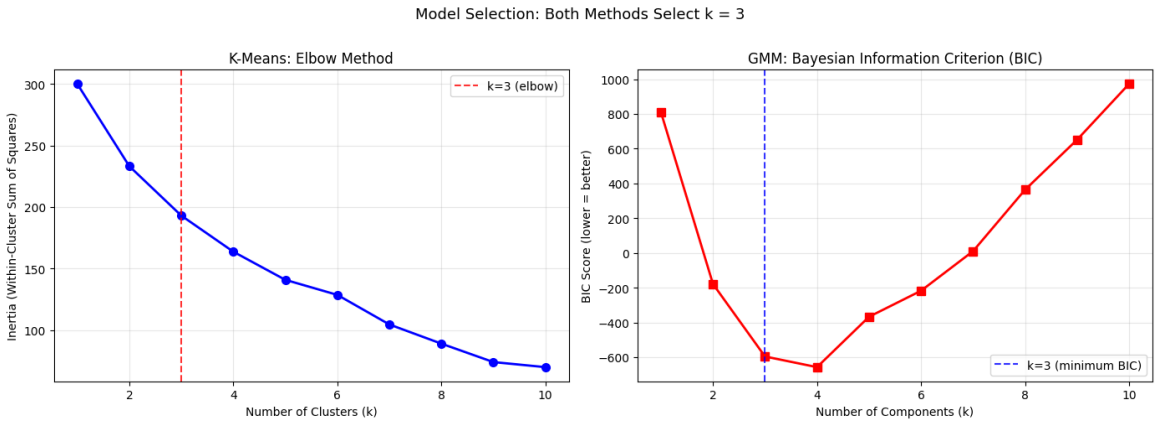
4.1 Model Selection: Elbow Method (K-Means) & BIC (GMM)

Both criteria point to **k = 3**:

- **K-Means:** inertia drops steeply from k=1 (300) to k=3 (193), then flattens — the "elbow" is at k=3
- **GMM:** BIC reaches its minimum at k=3 (−596), indicating the best balance of fit and model parsimony

In [19]:
```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))

# K-Means elbow plot
ax1.plot(range(1, 11), kmeans_scores, 'bo-', linewidth=2, markersiz
ax1.axvline(x=3, color='r', linestyle='--', alpha=0.8, label='k=3 (
ax1.set_title('K-Means: Elbow Method')
ax1.set_xlabel('Number of Clusters (k)')
ax1.set_ylabel('Inertia (Within-Cluster Sum of Squares)')
ax1.legend()
```

```
ax1.grid(True, alpha=0.3)

# GMM BIC plot
ax2.plot(range(1, 11), gmm_scores, 'rs-', linewidth=2, markersize=7
ax2.axvline(x=3, color='b', linestyle='--', alpha=0.8, label='k=3 (
ax2.set_title('GMM: Bayesian Information Criterion (BIC)')
ax2.set_xlabel('Number of Components (k)')
ax2.set_ylabel('BIC Score (lower = better)')
ax2.legend()
ax2.grid(True, alpha=0.3)

plt.suptitle('Model Selection: Both Methods Select k = 3', fontsize
plt.tight_layout()
plt.savefig('dist/figures/fig_elbow_bic.png', dpi=150, bbox_inches=
plt.show()
```



4.2 K-Means Clustering (k = 3)

We fit K-Means with k = 3 and print the team membership and centroid for each cluster.

In [20]:
```
k_best = 3

km = KMeans(n_clusters=k_best, init='k-means++', max_iter=200, n_in
km_labels = km.fit_predict(data_scaled)

print("=== K-Means Clusters (k=3) ===")
for c in range(k_best):
    teams = data_style.index[km_labels == c].tolist()
    print(f"Cluster {c}: {teams}")
```

```
=== K-Means Clusters (k=3) ===
Cluster 0: ['Arsenal', 'Aston Villa', 'Brighton', 'Chelsea', 'Fulha
m', 'Liverpool', 'Manchester City', 'Newcastle United', 'Tottenham H
otspur']
Cluster 1: ['Ipswich Town', 'Leicester City', 'Southampton', 'West H
am United']
Cluster 2: ['Bournemouth', 'Brentford', 'Crystal Palace', 'Everton',
'Manchester Utd', 'Nottingham Forest', 'Wolves']
```

4.3 Gaussian Mixture Model Clustering (k = 3)

GMM fits three multivariate Gaussian components via the EM algorithm. Cluster labels are assigned by the highest posterior probability (hard assignment from `predict`).

In [21]:
```
gm = GaussianMixture(n_components=k_best, random_state=123, n_init=
gm.fit(data_scaled)
gmm_labels = gm.predict(data_scaled)

print("=== GMM Clusters (k=3) ===")
for c in range(k_best):
    teams = data_style.index[gmm_labels == c].tolist()
    print(f"Cluster {c}: {teams}")
```

```
=== GMM Clusters (k=3) ===
Cluster 0: ['Brighton', 'Crystal Palace', 'Fulham', 'Manchester Ut
d', 'Tottenham Hotspur', 'Wolves']
Cluster 1: ['Bournemouth', 'Brentford', 'Everton', 'Ipswich Town', '
Leicester City', 'Nottingham Forest', 'Southampton', 'West Ham Unite
d']
Cluster 2: ['Arsenal', 'Aston Villa', 'Chelsea', 'Liverpool', 'Manch
ester City', 'Newcastle United']
```

4.4 GMM Soft Assignment Probabilities

GMM's theoretical advantage over K-Means is **soft assignment**: each team receives a probability for each cluster, not a hard label. With $n = 20$ and 15 features, however, we expect the posterior probabilities to become near-deterministic.

6. Conclusions

Both K-Means and GMM successfully identify **three playing style groups** within the 2024–25 EPL:

Style	K-Means (k=3)	GMM (k=3)
Attacking	9 teams (incl. Spurs, Brighton)	6 teams (core elite only)
Balanced	4 teams	6 teams (incl. Spurs, Brighton)
Defensive	7 teams (highest TklW)	8 teams (lowest poss/shots)

Key findings:

- A **core attacking group** (Arsenal, Aston Villa, Chelsea, Liverpool, Manchester City, Newcastle United) is robustly identified by both methods — high possession (≈56%), high shots (≈15/90), low tackles
- **Defensive-style teams** (K-Means: Bournemouth, Brentford etc.) win the most tackles per 90 (TklW ≈ 11.4), reflecting a low-block or counter-pressing approach

- The main divergence is Tottenham and Brighton — K-Means places them in the attacking group (high possession), GMM places them in the balanced group (below-elite attacking output)
- **GMM soft assignments** collapsed to hard 0/1 due to the small sample (20 teams, 15 features), negating the probabilistic advantage

See `dist/report.tex` for the full academic write-up and references.

In [22]:
```
probs = gm.predict_proba(data_scaled)

probs_df = pd.DataFrame(
    probs,
    index=data_style.index,
    columns=[f'Cluster {i}' for i in range(k_best)]
).round(3)

print("=== GMM Cluster Probabilities ===")
probs_df
```

```
=== GMM Cluster Probabilities ===
```

Out[22]:

Squad	Cluster 0	Cluster 1	Cluster 2
Arsenal	0.0	0.0	1.0
Aston Villa	0.0	0.0	1.0
Bournemouth	0.0	1.0	0.0
Brentford	0.0	1.0	0.0
Brighton	1.0	0.0	0.0
Chelsea	0.0	0.0	1.0
Crystal Palace	1.0	0.0	0.0
Everton	0.0	1.0	0.0
Fulham	1.0	0.0	0.0
Ipswich Town	0.0	1.0	0.0
Leicester City	0.0	1.0	0.0
Liverpool	0.0	0.0	1.0
Manchester City	0.0	0.0	1.0
Manchester Utd	1.0	0.0	0.0
Newcastle United	0.0	0.0	1.0
Nottingham Forest	0.0	1.0	0.0
Southampton	0.0	1.0	0.0
Tottenham Hotspur	1.0	0.0	0.0
West Ham United	0.0	1.0	0.0
Wolves	1.0	0.0	0.0

4.5 Detailed Cluster Analysis (Helper Functions)

The helper functions below refit each model and print full cluster membership and centroid statistics for any value of k. We use them here to confirm the k = 4 results are less interpretable than k = 3.

In [23]:
```
def list_clusters_kmeans(data_scaled, data_style, n_clusters):
    model = KMeans(n_clusters=n_clusters, init='k-means++', max_ite
                   n_init=10, random_state=0)
    model.fit(data_scaled)
    labels = model.labels_

    print(f'\nThere are {n_clusters} clusters with inertia score of

    for label in range(n_clusters):
```

```
        teams = data_style.index[labels == label].tolist()
        centre = data_style.loc[teams].mean().round(2).to_dict()
        print(f'Cluster {label} has {len(teams)} teams: {teams}')
        print(f'Centre: {centre}\n')

    return labels

def list_clusters_gmm(data_scaled, data_style, n_clusters):
    model = GaussianMixture(n_components=n_clusters, random_state=1
    model.fit(data_scaled)
    labels = model.predict(data_scaled)
    score = model.bic(data_scaled)

    print(f'\nThere are {n_clusters} clusters with BIC score of {sc

    for label in range(n_clusters):
        teams = data_style.index[labels == label].tolist()
        centre = data_style.loc[teams].mean().round(2).to_dict()
        print(f'Cluster {label} has {len(teams)} teams: {teams}')
        print(f'Centre: {centre}\n')

    return labels
```

4.6 Sensitivity Analysis: Exploring k = 4

We run both methods with k = 4 to verify that k = 3 is a robust choice. At k = 4, K-Means isolates Brentford as a singleton cluster, indicating it is an outlier rather than a genuinely distinct style group. GMM at k = 4 splits the dominant group but produces less interpretable boundaries. This confirms k = 3 as the most meaningful solution.

In [24]:
```
print("=== K-Means Cluster Analysis (k=4) ===")
km_labels = list_clusters_kmeans(data_scaled, data_style, 4)
```

```
=== K-Means Cluster Analysis (k=4) ===

There are 4 clusters with inertia score of 163.64

Cluster 0 has 1 teams: ['Brentford']
Centre: {'Poss': 47.9, 'CrdY': 1.63, 'CrdR': 0.03, 'Sh': 11.4, 'So
T': 4.55, 'SoT%': 39.9, 'Fls': 8.42, 'Fld': 8.92, 'Off': 1.66, 'Cr
s': 17.7, 'Int': 7.5, 'TklW': 10.0, 'SoTA': 5.45, 'Saves': 4.03, 'Sa
ve%': 72.9}

Cluster 1 has 10 teams: ['Bournemouth', 'Brighton', 'Crystal Palac
e', 'Everton', 'Fulham', 'Manchester Utd', 'Nottingham Forest', 'Tot
tenham Hotspur', 'West Ham United', 'Wolves']
Centre: {'Poss': 48.27, 'CrdY': 2.17, 'CrdR': 0.07, 'Sh': 12.92, 'So
T': 4.36, 'SoT%': 33.74, 'Fls': 11.62, 'Fld': 10.65, 'Off': 1.77, 'C
rs': 18.14, 'Int': 8.63, 'TklW': 11.33, 'SoTA': 4.31, 'Saves': 2.9,
'Save%': 68.59}

Cluster 2 has 3 teams: ['Ipswich Town', 'Leicester City', 'Southampt
on']
Centre: {'Poss': 44.83, 'CrdY': 2.36, 'CrdR': 0.07, 'Sh': 9.18, 'So
T': 2.9, 'SoT%': 31.6, 'Fls': 11.5, 'Fld': 10.67, 'Off': 1.44, 'Cr
s': 14.67, 'Int': 8.2, 'TklW': 10.07, 'SoTA': 5.83, 'Saves': 3.67, '
Save%': 64.93}

Cluster 3 has 6 teams: ['Arsenal', 'Aston Villa', 'Chelsea', 'Liverp
ool', 'Manchester City', 'Newcastle United']
Centre: {'Poss': 55.8, 'CrdY': 1.93, 'CrdR': 0.08, 'Sh': 14.8, 'So
T': 4.94, 'SoT%': 33.32, 'Fls': 10.34, 'Fld': 11.14, 'Off': 1.65, 'C
rs': 18.1, 'Int': 6.74, 'TklW': 9.44, 'SoTA': 3.77, 'Saves': 2.63, '
Save%': 71.58}
```

In [25]:

```python
print("=== GMM Cluster Analysis (k=4) ===")
gmm_labels = list_clusters_gmm(data_scaled, data_style, 4)
```

```
=== GMM Cluster Analysis (k=4) ===

There are 4 clusters with BIC score of -657.32

Cluster 0 has 6 teams: ['Arsenal', 'Aston Villa', 'Chelsea', 'Manche
ster City', 'Newcastle United', 'Tottenham Hotspur']
Centre: {'Poss': 55.3, 'CrdY': 1.96, 'CrdR': 0.07, 'Sh': 14.17, 'So
T': 4.77, 'SoT%': 33.7, 'Fls': 10.41, 'Fld': 11.68, 'Off': 1.66, 'Cr
s': 18.2, 'Int': 6.84, 'TklW': 9.54, 'SoTA': 3.95, 'Saves': 2.72, 'S
ave%': 70.68}

Cluster 1 has 5 teams: ['Bournemouth', 'Brentford', 'Brighton', 'Ful
ham', 'Liverpool']
Centre: {'Poss': 51.74, 'CrdY': 2.02, 'CrdR': 0.06, 'Sh': 14.14, 'So
T': 4.89, 'SoT%': 34.82, 'Fls': 11.28, 'Fld': 9.82, 'Off': 1.48, 'Cr
s': 20.16, 'Int': 8.17, 'TklW': 10.58, 'SoTA': 4.33, 'Saves': 3.02,
'Save%': 70.68}

Cluster 2 has 5 teams: ['Crystal Palace', 'Everton', 'Manchester Ut
d', 'Nottingham Forest', 'Wolves']
Centre: {'Poss': 45.3, 'CrdY': 2.2, 'CrdR': 0.07, 'Sh': 12.26, 'So
T': 4.21, 'SoT%': 34.38, 'Fls': 11.22, 'Fld': 10.53, 'Off': 1.98, 'C
rs': 16.32, 'Int': 8.72, 'TklW': 11.92, 'SoTA': 4.18, 'Saves': 2.81,
'Save%': 68.5}

Cluster 3 has 4 teams: ['Ipswich Town', 'Leicester City', 'Southampt
on', 'West Ham United']
Centre: {'Poss': 45.72, 'CrdY': 2.31, 'CrdR': 0.07, 'Sh': 9.99, 'So
T': 3.08, 'SoT%': 30.95, 'Fls': 11.55, 'Fld': 10.6, 'Off': 1.56, 'Cr
s': 15.02, 'Int': 8.33, 'TklW': 10.1, 'SoTA': 5.59, 'Saves': 3.58, '
Save%': 65.78}
```

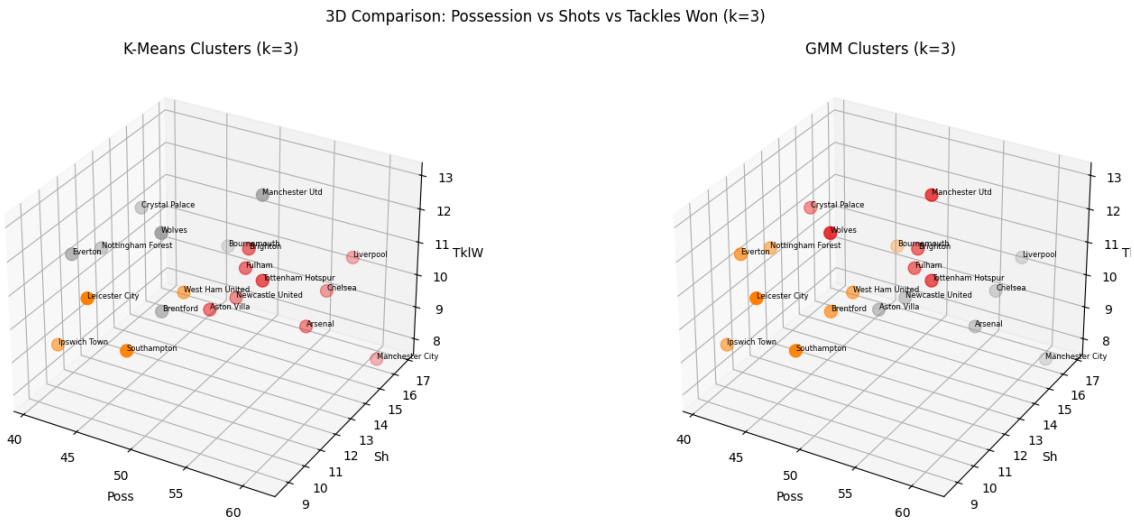
5. Comparison and Visualisation

Side-by-side 3D scatter plot comparing K-Means and GMM cluster assignments across three key style dimensions: **Possession**, **Shots**, and **Tackles Won**.

In [26]:

```
from mpl_toolkits.mplot3d import Axes3D

# Recompute k=3 labels explicitly for this visualisation
# (km_labels and gmm_labels were overwritten by the k=4 analysis ab
km_vis = KMeans(n_clusters=k_best, init='k-means++', max_iter=200,
km_labels_vis = km_vis.fit_predict(data_scaled)

gm_vis = GaussianMixture(n_components=k_best, random_state=123, n_i
gm_vis.fit(data_scaled)
gmm_labels_vis = gm_vis.predict(data_scaled)

# 3D scatter: Possession vs Shots vs Tackles Won — key style dimens
x_axis = 'Poss'
y_axis = 'Sh'
z_axis = 'TklW'
```

```
fig = plt.figure(figsize=(16, 6))

# K-Means k=3 — 3D scatter
ax1 = fig.add_subplot(121, projection='3d')
ax1.scatter(data_style[x_axis], data_style[y_axis], data_style[z_ax
            c=km_labels_vis, cmap=plt.colormaps['Set1'], s=100)
for i, team in enumerate(data_style.index):
    ax1.text(data_style[x_axis].iloc[i], data_style[y_axis].iloc[i]
             data_style[z_axis].iloc[i], team, fontsize=6)
ax1.set_xlabel(x_axis)
ax1.set_ylabel(y_axis)
ax1.set_zlabel(z_axis)
ax1.set_title(f'K-Means Clusters (k={k_best})')

# GMM k=3 — 3D scatter
ax2 = fig.add_subplot(122, projection='3d')
ax2.scatter(data_style[x_axis], data_style[y_axis], data_style[z_ax
            c=gmm_labels_vis, cmap=plt.colormaps['Set1'], s=100)
for i, team in enumerate(data_style.index):
    ax2.text(data_style[x_axis].iloc[i], data_style[y_axis].iloc[i]
             data_style[z_axis].iloc[i], team, fontsize=6)
ax2.set_xlabel(x_axis)
ax2.set_ylabel(y_axis)
ax2.set_zlabel(z_axis)
ax2.set_title(f'GMM Clusters (k={k_best})')

plt.suptitle('3D Comparison: Possession vs Shots vs Tackles Won (k=
plt.tight_layout()
plt.savefig('dist/figures/fig_3d_scatter.png', dpi=150, bbox_inches
plt.show()
```



6. Conclusions

Both K-Means and GMM successfully identify **three meaningful playing style groups** within the 2024–25 EPL:

Cluster	K-Means (k=3)	GMM (k=3)

Elite / Dominant	9 teams (incl. Spurs, Brighton)	6 teams (core elite only)
Mid-table / Resilient	7 teams	6 teams
Struggling / Lower-half	4 teams	8 teams

Key findings:

- A **core elite group** (Arsenal, Aston Villa, Chelsea, Liverpool, Manchester City, Newcastle United) is robustly identified by both methods
- The main divergence is Tottenham and Brighton — high possession but below elite in attacking output — placed in the dominant group by K-Means and the mid-table group by GMM
- **GMM soft assignments** collapsed to hard 0/1 due to the small sample (20 teams, 15 features), negating the probabilistic advantage
- Cluster profiles align with tactical intuition: elite sides dominate possession and shots; struggling sides have low shot volumes and face many shots against

See `dist/report.tex` for the full academic write-up and references.